Uniformly Lipschitz Sequences: Pointwise Convergence Implies Uniform Convergence

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Problem Statement. Let (f_n) be a sequence of functions on $[a, b]$ with a common Lipschitz constant L, so that

$$|f_n(x) - f_n(y)| \leq L|x - y| \qquad \text{for all } n \in \mathbb{N},\ x, y \in [a, b].$$

(1) If $f = \lim_n f_n$ pointwise, then f is continuous and satisfies a Lipschitz condition with constant L.

(2) If $f = \lim_n f_n$ pointwise, then the convergence is uniform.

(3) The conclusions of (1) and (2) can fail if each f_n is only assumed to be Lipschitz, with a constant that may depend on n.

Part (1)

Proof. Fix $x, y \in [a, b]$. For every n,

$$|f_n(x) - f_n(y)| \leq L|x - y|.$$

Since $f_n(x) \to f(x)$ and $f_n(y) \to f(y)$, we have $f_n(x) - f_n(y) \to f(x) - f(y)$. The absolute value is continuous, so $|f_n(x) - f_n(y)| \to |f(x) - f(y)|$. Non-strict inequalities are preserved under limits, so letting $n \to \infty$ gives

$$|f(x) - f(y)| \leq L|x - y|.$$

Thus f is Lipschitz with constant L.

Every Lipschitz function is continuous. Indeed, if $L = 0$ then f is constant. If $L > 0$, then given $\varepsilon > 0$, let $\delta = \varepsilon/L$. Whenever $|x - y| < \delta$, we get $|f(x) - f(y)| \leq L|x - y| < \varepsilon$. (In fact f is uniformly continuous on $[a, b]$.) □

Part (2)

The idea is that pointwise convergence at finitely many points is automatically uniform over those points, and the common Lipschitz constant controls every f_n, and f itself, between neighboring points.

Proof. Let $\varepsilon > 0$. By Part (1), f is also Lipschitz with constant L.

Choosing a finite grid. Let $\delta = \dfrac{\varepsilon}{3L+1} > 0$, so that $L\delta < \dfrac{\varepsilon}{3}$. Choose $k \in \mathbb{N}$ with $\dfrac{b-a}{k} < \delta$, and set

$$x_i = a + i \cdot \frac{b-a}{k}, \qquad i = 0, 1, \ldots, k.$$

Every $x \in [a, b]$ lies in some interval $[x_{i-1}, x_i]$, so there is an index i with $|x - x_i| \leq \dfrac{b-a}{k} < \delta$.

Using pointwise convergence on the grid. For each i, since $f_n(x_i) \to f(x_i)$, there exists N_i such that $|f_n(x_i) - f(x_i)| < \dfrac{\varepsilon}{3}$ for all $n \geq N_i$. Let $N = \max\{N_0, N_1, \ldots, N_k\}$. Then for all $n \geq N$ and all i,

$$|f_n(x_i) - f(x_i)| < \frac{\varepsilon}{3}.$$

Estimating at an arbitrary point. Let $n \geq N$ and $x \in [a, b]$, and choose i with $|x - x_i| < \delta$. By the triangle inequality,

$$\begin{aligned} |f_n(x) - f(x)| &\leq |f_n(x) - f_n(x_i)| + |f_n(x_i) - f(x_i)| + |f(x_i) - f(x)| \\ &< L\delta + \frac{\varepsilon}{3} + L\delta \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

The first and third terms are bounded using the Lipschitz conditions on f_n and f, both with constant L.

Since N does not depend on x, we have $\sup_{x \in [a,b]} |f_n(x) - f(x)| \leq \varepsilon$ for all $n \geq N$. Hence $f_n \to f$ uniformly on $[a, b]$. □

Part (3)

A single example shows that both conclusions fail. On $[a, b]$, define

$$f_n(x) = \left(\frac{x-a}{b-a}\right)^n.$$

To simplify notation, let $u = \frac{x-a}{b-a}$, which ranges over $[0, 1]$ as x ranges over $[a, b]$.

Each f_n is Lipschitz. For $u, v \in [0, 1]$,

$$|u^n - v^n| = |u - v| \cdot \left|u^{n-1} + u^{n-2}v + \cdots + v^{n-1}\right| \leq n\,|u - v|,$$

since each of the n terms in the second factor lies in $[0, 1]$. Therefore, for $x, y \in [a, b]$,

$$|f_n(x) - f_n(y)| \leq n \left|\frac{x-a}{b-a} - \frac{y-a}{b-a}\right| = \frac{n}{b-a}\,|x - y|.$$

So f_n is Lipschitz with constant $L_n = \frac{n}{b-a}$, which grows with n.

The pointwise limit. If $a \leq x < b$, then $0 \leq u < 1$, so $u^n \to 0$. If $x = b$, then $u = 1$ and $u^n = 1$ for all n. Hence $f_n \to f$ pointwise, where

$$f(x) = \begin{cases} 0 & \text{if } a \leq x < b, \\ 1 & \text{if } x = b. \end{cases}$$

Conclusion (1) fails. The limit f is discontinuous at b, since $\lim_{x\to b^-} f(x) = 0 \neq 1 = f(b)$. In particular, f is not Lipschitz with any constant.

Conclusion (2) fails. Each f_n is continuous. If the convergence were uniform, the limit f would be continuous, since a uniform limit of continuous functions is continuous. This contradicts the previous paragraph. One can also see it directly: for each fixed n, $f_n(x) \to 1$ as $x \to b^-$ while $f(x) = 0$ for $x < b$, so

$$\sup_{x\in[a,b]} |f_n(x) - f(x)| \geq \sup_{a\leq x<b} f_n(x) = 1 \qquad \text{for every } n,$$

which does not tend to 0.

Hence, without a common Lipschitz constant, the pointwise limit need not be continuous, and the convergence need not be uniform. □